# Supplementary Methods and Results

Decision-Task Portability of PV Characterization Sequences: A Retrospective Cross-Climate Audit

*This supplement provides the detailed source admission, fixed-sequence replay, task-specific stopping-depth audits, exact-unit screening intervals and calibration sensitivities underlying the separately defined decision operations.* **Public derived-only reproducibility.** The frozen

190-run task-comparison ledger, validity verifier, and the 12-case Table S14 source are available at https://github.com/shaikhamalkawi-ux/pv-evidence-portability-across-climates/tree/cb42962c6544a5f9afb4e69d39682799cf932370/reproducibility/R18. The repository does not redistribute original producer measurements or establish a prospective measurement certificate.

## 1 Acquisition-order branch: data filtering and response geometry

The acquisition branch uses the definitions established in the main manuscript. The response panel is the Sandia PV Performance Modeling Collaborative 2021 blind-comparison IEC 61853-1 matrix archive for modules 19074-001 through 19074-009; the retained analysis file has nine products by 27 nodes. This response source is distinct from the NREL mPERT field records used to construct Cocoa, Eugene and Golden operating-state weights. The NREL core contains five field technologies in each of those three sites. Across the 15 field files, 462,429 records pass the supplied finite-value filters before interpolation admission. Field states outside the fixed response-grid convex hull are excluded rather than extrapolated. Site weights average the five technology-specific basis-weight vectors equally.

The Switzerland Task 13 outdoor series contains 36,449 five-minute rows from July 2018 through June 2019. Of these, 34,763 rows satisfy the declared finite/positive conditions and 28,292 (81.385%) lie inside the fixed triangulation. Albuquerque contains 105,409 five-minute bins; 49,939 satisfy the finite-temperature/daylight rule and 36,911 (73.912%) lie inside the triangulation. The raw Task 13 monthly files and Albuquerque NetCDF source are identified by the hashes recorded in the author-controlled complete replay archive.

## 2 Complete directed characterization-order matrix

Table S1 provides the full source-by-target values underlying the main-text heatmap.

**Table S1:** Five-site directed excess characterization depth (%). Rows are source-conditioned orders; columns are target environments.

| Source | Cocoa | Eugene | Golden | Switzerland | Albuquerque |
|---|---|---|---|---|---|
| Cocoa | 0.00 | 8.25 | 10.03 | 14.86 | 21.43 |
| Eugene | 45.95 | 0.00 | 57.75 | 60.47 | 68.83 |
| Golden | -1.69 | 2.06 | 0.00 | 2.36 | 4.22 |
| Switzerland | 2.36 | -0.34 | 3.04 | 0.00 | 6.49 |
| Albuquerque | 1.35 | -1.37 | 0.61 | -1.35 | 0.00 |

All 25 aggregate cells resolve all 36 held response-panel pairs with zero wrong accepted stopping decisions. Figure S1 summarizes the corresponding source-level off-target burden and worst target.

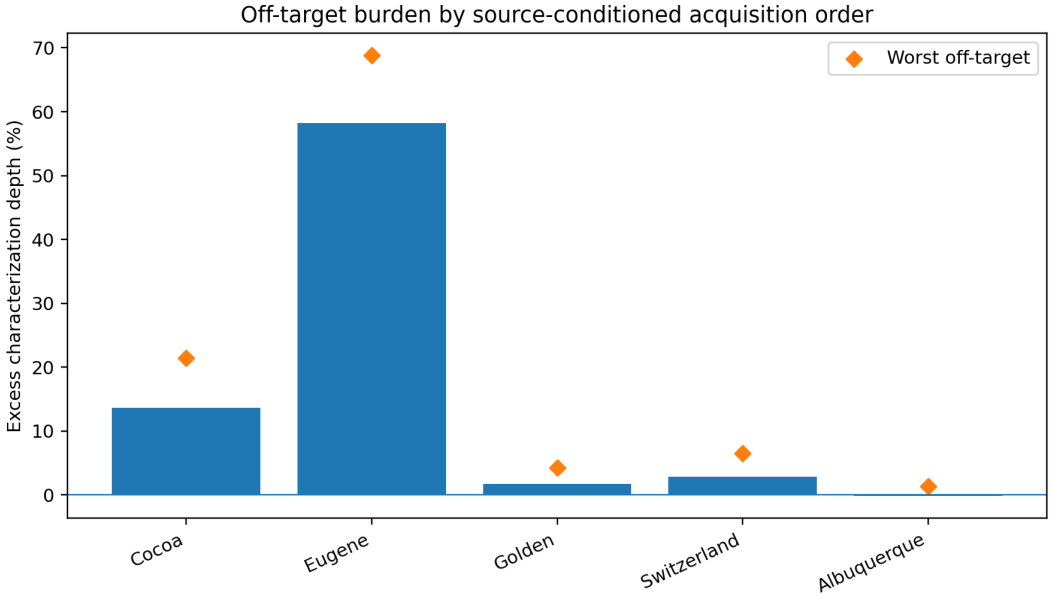



**Figure S1:** Mean off-target excess burden by source-conditioned acquisition order; diamonds show the worst off-target value.

## 3 Five-site common support and common-order comparators

The five-environment intersection contains 27 of 35 triangulation simplexes. Retention relative to already-in-grid records is 99.879% in Cocoa, 99.171% in Eugene, 99.230% in Golden, 99.526% in Switzerland and 98.057% in Albuquerque. On this support, source mean/worst excess values are: Albuquerque -1.00/0.00%, Golden +1.65/+5.25%, Switzerland +2.64/+7.54%, Cocoa +13.17/+21.97%, and Eugene +58.24/+69.51%.

The equal-site GlobalPooled5 order is retrospective because every target contributes to its construction. Its target excesses are +3.04% (Cocoa), -1.72% (Eugene), +7.60% (Golden), +7.77% (Switzerland) and +11.36% (Albuquerque). The target-excluded LOCO5 values are +8.78%, +4.81%, +10.03%, +19.59% and +25.65%, respectively. These are common-order comparators and do not establish optimality.

## 4 Technology-stratified and random-order sensitivities

Re-evaluation on the 15 individual U.S.-core technology trajectories gives mean excess values of -0.79% (Albuquerque), -0.77% (Golden), +1.24% (Switzerland), +4.47% (Cocoa), and +31.32% (Eugene). Albuquerque, Golden and Cocoa each produce one wrong accepted decision in one technology-specific trajectory; Eugene has one unresolved trajectory. This prevents a universal claim that any low-burden source order is sign-preserving for every field technology.

One thousand random permutations are evaluated per off-target transfer using the same source-admissible node set. The Albuquerque and Golden source orders occupy the extreme low-burden tail of the matched random reference across the evaluated targets. The random-order analysis provides a scale reference; it does not establish that the deterministic source orders are statistically optimal policies.

## 5 Why symmetric climate distance does not reconstruct directed burden

For any symmetric distance $d(s,t) = d(t,s)$ and any scalar function $g$, $g(d(s,t)) = g(d(t,s))$. If the directed burden of the first transfer differs from its reverse, i.e. $E^{\mathrm{acq}}_{s\to t} \neq E^{\mathrm{acq}}_{t\to s}$, no predictor depending *only* on that symmetric distance can fit both directions exactly. This is an elementary algebraic constraint, not a novel physical theorem or a claim that symmetric distances are uninformative. They can still summarize unordered-pair difficulty; characterization-order transfer is direction-dependent.

## 6 Operating-state distance diagnostic

Across the ten undirected environment pairs, Spearman correlations between symmetric distance and mean bidirectional excess are approximately 0.855 for Jensen–Shannon and Hellinger distance, 0.818

for normalized two-dimensional Wasserstein-1, and 0.794 for total variation. The sample contains only ten pairs and the diagnostic was developed after the directional result, so nominal p-values are not confirmatory. More importantly, each distance is symmetric and cannot explain large source-to-target directional gaps.

# 7 Shortlist branch: complete Cocoa–Eugene exact-unit panel

Table S2 records the complete exact-unit endpoint used to construct all Cocoa–Eugene shortlist comparisons. The source annual windows are March 2011–February 2012 at Cocoa and January–December 2013 at Eugene, using the same physical units redeployed sequentially rather than simultaneously. The source manual reports measurement history and time-varying performance diagnostics, including sensitivity to spectral conditions for some thin-film modules. These data support nominal within-site deficits and directed candidate sets but do not separately identify climatic and deployment-time effects.

**Table S2:** Complete exact-unit annual final-yield panel and deficits from the site-best value.

| Module | Label | Cocoa $Y_f$/deficit | Eugene $Y_f$/deficit |
|---|---|---|---|
| CIGS39017 | CIGS-1 | 1740 h / 13.819% | 1344 h / 9.312% |
| CIGS8-001 | CIGS-2 | 1716 h / 15.007% | 1270 h / 14.305% |
| CdTe75638 | CdTe | 1986 h / 1.634% | 1482 h / 0.000% |
| HIT05667 | HIT | 1855 h / 8.123% | 1434 h / 3.239% |
| aSiTandem72-46 | a-Si tandem | 1917 h / 5.052% | 1430 h / 3.509% |
| aSiTriple28324 | a-Si triple | 1904 h / 5.696% | 1436 h / 3.104% |
| aSiMicro03036 | a-Si/$\mu$c-Si | 2019 h / 0.000% | 1482 h / 0.000% |
| mSi0166 | mc-Si-1 | 1762 h / 12.729% | 1362 h / 8.097% |
| mSi0188 | mc-Si-2 | 1769 h / 12.382% | 1378 h / 7.018% |
| mSi460A8 | mc-Si-3 | 1806 h / 10.550% | 1403 h / 5.331% |
| xSi12922 | x-Si | 1800 h / 10.847% | 1406 h / 5.128% |

# 8 All Cocoa-to-Eugene exact-unit false-exclusion intervals

Table S3 reports every nonempty exact-unit interval; the main text summarizes their union and maximum simultaneous exclusion count.

**Table S3:** Nonempty Cocoa→Eugene exact-unit false-exclusion intervals.

| Module | Label | Low (%) | High (%) | Width (pp) |
|---|---|---|---|---|
| CdTe75638 | CdTe | 0.000 | 1.634 | 1.634 |
| aSiTriple28324 | a-Si triple | 3.104 | 5.696 | 2.592 |
| HIT05667 | HIT | 3.239 | 8.123 | 4.884 |
| aSiTandem72-46 | a-Si tandem | 3.509 | 5.052 | 1.543 |
| xSi12922 | x-Si | 5.128 | 10.847 | 5.719 |
| mSi460A8 | mc-Si-3 | 5.331 | 10.550 | 5.219 |
| mSi0188 | mc-Si-2 | 7.018 | 12.382 | 5.365 |
| mSi0166 | mc-Si-1 | 8.097 | 12.729 | 4.632 |
| CIGS39017 | CIGS-1 | 9.312 | 13.819 | 4.507 |
| CIGS8-001 | CIGS-2 | 14.305 | 15.007 | 0.702 |

The reverse Eugene→Cocoa direction has no nonempty false-exclusion interval in this annual endpoint. At $q = 5\%$, Cocoa→Eugene has three false exclusions, zero excess retentions and 40% target-set coverage; Eugene→Cocoa has zero false exclusions and three excess retentions.

**Source-reported measurement uncertainty versus nominal screening.** Marion et al. (2014) report an expanded uncertainty ($k = 2$) of 3.5% for final PV yield $Y_f$ in Table 3 and approximately 5% uncertainty when comparing two annual yields using their stated root-sum-of-squares combination. This is existing source evidence, not a new uncertainty estimate. In this deterministic screening audit, $q = 5\%$ is a tolerance around the observed site-best value, *not* an uncertainty interval or a statistical significance level. For the three nominally excluded Eugene-relevant units, the Cocoa deficits exceed this cutoff by 0.052 percentage points (aSiTandem72-46), 0.696 (aSiTriple28324), and 3.123 (HIT05667). A valid uncertainty interval for their normalized $\delta$ values would additionally require the covariance structure

between yields and the uncertainty introduced by selecting the site-best denominator. Neither the 3.5% nor the comparison-level approximately 5% may be copied directly as a *plus/minus five-percentage-point* uncertainty band on $\delta$. Therefore the interval membership records in Tables S2–S3 describe nominal estimates; no source-to-target exclusion receives a calibrated robustness label.

## 9 External Oslo–Tozeur–Sevilla preprocessing and sanity check

The external source is Zenodo v2, doi: 10.5281/zenodo.10423357. The analyzed `PV_data.zip` is 374,924,215 bytes with MD5 `6e1cae3a4428a79ace289a3f0e187b02`; its full SHA-256 is retained in the author-controlled source manifest. Cadence inspection identifies a modal positive 60-s spacing in the raw records at all three sites. The December 2021–November 2023 archive is grouped into five-minute bins without interpolation. Five-minute module power is normalized by public STC nameplate power, then averaged within configuration when at least 75% of assigned modules are positive and finite. Site timestamps require all seven configurations. The primary common calendar contains 23 months.

The reconstructed M1 bifacial-versus-monofacial differences are 9.995% in Oslo, 21.464% in Tozeur and 8.949% in Sevilla, closely matching the independent study's rounded values of approximately 10%, 21% and 8%. This is a processing sanity check only.

## 10 External deficits, intervals and robustness

M1-HFL, M1-HFB and M1-Bi are source-archive configuration identifiers, not mathematical quantities or serial identities of the nine Stage-A products. At $q = 5\%$, Oslo retains M1-HFL and M1-HFB; Tozeur and Sevilla additionally retain M1-Bi. The M1-Bi deficits are 10.542%, 2.828% and 1.145%, respectively. The primary false-exclusion intervals are therefore [2.828,10.542)% for Oslo→Tozeur and [1.145,10.542)% for Oslo→Sevilla.

The result is unchanged at 50%, 75% and 100% within-type coverage. Splitting the 23 common months into an 11-month first block and a 12-month second block also preserves the same 5% shortlists. Finally, 10,000 paired-month bootstrap resamples preserve the M1-Bi exclusion in 10,000/10,000 resamples for both Oslo→Tozeur and Oslo→Sevilla. The descriptive 2.5/50/97.5 percentiles of interval width are 5.724/7.687/10.744 pp and 7.264/9.371/12.667 pp, respectively.

## 11 Independent Stage-A field-source identity and admission

The external Stage-A targets are sourced from Safi (Mendeley Data v1, doi: 10.17632/b2s4ycc2gz.1), Germany MetPVNet (Zenodo 3958820), the Trieste Q2 array (Zenodo 7115550), and BR-PVGen (Zenodo 21511487). Published CC BY 4.0 licenses and original MD5 digests were checked for the three Zenodo releases; the separately supplied original Safi ZIP was CRC and SHA-256 checked but a publisher-side Safi ZIP SHA is not available through the blocked Mendeley API. The Safi ZIP includes seven unedited source workbooks and a preprocessed workbook that duplicates a separately shared XLSX byte for byte. The preprocessed workbook is an additional *sensitivity* dataset, not another independent field deployment.

The frozen original response-grid support gate retains only finite positive measured POA and finite measured module temperature, with no beyond-hull extrapolation. Numeric retention is conditional on *simultaneously paired daylight readings*, not on full nocturnal grid entries or missing nights. Each historical order is evaluated unchanged against a target-local order derived under the original fixed rule; the target-local order is not a mathematically certified optimum. Table S4 distinguishes paired daylight readings from observations inside the unchanged response grid, making source-specific admission losses explicit.

**Table S4:** Independent Stage-A source admission denominators, conditional on finite paired daylight readings. Brazil aggregates 34 eligible plant records, *not* 34 independent climates; the remaining 17 plant files do not satisfy the admission criteria. The one-minute German series and 15-minute Italy/Brazil series are source-resampled or aggregated.

| Source trajectory | Paired valid $G$–$T$ | In 27-node grid | Retained (%) | Observed dates |
|---|---|---|---|---|
| Germany 2018 / S1 | 30,605 | 19,937 | 65.1 | 40 |
| Germany 2018 / S2A | 28,310 | 19,800 | 69.9 | 40 |
| Germany 2019 / S2B | 32,370 | 21,069 | 65.1 | 37 |
| Italy Q2/inv1 | 151,211 | 86,859 | 57.4 | 2689 |
| Brazil 34 plant totals | 398,394 | 269,858 | 67.7 | – |

For Safi, a source-clock nearest-neighbor join within two seconds (without interpolation) yields 42,332/98,774 (42.86%), 13,197/32,728 (40.32%) and 26,194/59,524 (44.01%) within-grid pairs for m-Si, p-Si and a-Si, respectively. Raw logger windows are unequal; each series is analyzed as a separate seasonal/technology conditioning environment from *one* Moroccan PV site. The 60-second synchronization and source-provided preprocessed table are kept separate: one preprocessing sensitivity changes the preferred historical order in m-Si, and several source features align to lagged original readings. Neither check supplies a metrological uncertainty model.

## 12 Independent replay and source integrity

The unmodified archived baseline calibration code, source weights for five original environments, response matrix, 36 held pair comparisons, seven-product residual envelope, strict stopping inequality and STC counting were restored. Against the immutable reference 5×5 matrix, all 25 source–target transfer values and original wrong/unresolved flags match at tolerance $10^{-8}$ percentage points. The archived replay provides outputs for 150 supported target *trajectories*, of which only 38 are primary: three German system-periods, one Italian full-period sensor and 34 distinct Brazilian plant files. The other 112 are repeated Italy sensors (2), Italy years (9), and Brazilian calendar-quarter slices (101). All are source-correlated exploratory sensitivities, not independent site-level replications.

The author-controlled derived archive contains 900 target–source comparisons (150 targets × six orders including local) and 32,400 pair-level decisions (900 × 36). A separately implemented output verifier reconstructs 900 means, all valid excess percentages, 36-pair row counts, near-tie flags, and unresolved/correctness totals; 26 independent output checks pass. The complete source identities, raw-input replay and exact comparison results are described below. There is no stochastic metrological confidence interval or proxy-to-laboratory-time conversion.

## 13 Raw-input replay and measurement prerequisites

**Fresh replay.** A fresh execution used the archived SHA-256-verified Stage-A program and the retained Swiss, Albuquerque, German, Italian and Brazilian producer files. The six German/Italian/Brazilian input-file hashes matched their frozen source manifest. The original five-site transfer matrix passed 25/25 comparisons before the external replay. The regenerated 230 admission rows, 150 target-weight vectors, 900 order–target rows and 32,400 pair decisions matched the archived reference CSVs in row order, columns, text and numerics (maximum numerical discrepancy 0); an independently implemented output verifier passed 26/26 checks. This is same-code raw-input reproduction, not independent algorithm development or source-archive redistribution.

**Fixed decision procedure.** For each target, accept only $G_{\mathrm{POA}}$–$T_{\mathrm{mod}}$ states inside the 27-node grid and form nonnegative barycentric weights. A historical source order visits non-STC nodes with weight $> 10^{-14}$ in descending source-weight order, resolving ties by irradiance and temperature. For each held pair, the other seven complete 27-node product profiles provide 21 calibration-pair residuals. At each visited prefix, compare the held pair's partial weighted contrast with the maximum absolute full-versus-prefix calibration residual; stop only when the strict inequality in Eq. (3) of the Main holds. STC is mandatory. Afterward, compare the accepted sign to the full-reference contrast; otherwise record unresolved. Aggregate mean depth, $K_{\mathrm{all}}$ and $N_{\mathrm{mod-cell}}$ only with appropriate complete-run denominators.

**Feasibility and uncertainty.** Seven fully measured calibration profiles are a prerequisite per held pair. Rotating across all 36 pairs presupposes the characterized nine-product panel; its acquisition cost is

unmeasured. Physical switching, stabilization, repeated scans and calibrated cell-level uncertainties are not supplied. No synthetic perturbation range is admitted as measurement uncertainty; the observed near-tie decision states cannot be converted to calibrated risk probabilities. Public GitHub contains derived checks rather than the full raw-input archive.

## 14 Information availability in Stage-A stopping

Each of 36 evaluated pairs leaves its two products out of the calibration set. The seven *other* complete module profiles supply $\binom{7}{2} = 21$ calibration-pair residuals. Before held-pair testing, the target $G_{\mathrm{POA}}$–$T_{\mathrm{mod}}$ weights and support, a fixed source order and those seven fully characterized calibrators are known. During testing, only STC normalization and the held pair's partial responses at visited non-STC nodes enter the strict stop rule. An unresolved outcome is known after all eligible nodes are exhausted. The full held-pair contrast and correct/wrong accepted-sign labels are available *only afterward* for retrospective validation.

An independent disclosure-restricted replay reproduced all *5,400/5,400* archived target-local stop depths, unresolved flags and post-hoc wrong-sign flags across 150 supported target weight vectors. The full held-pair contrast is not an argument of the stopping function. This is a computational information-flow check on the *same* nine response matrices, not a new physical experiment, metrological coverage guarantee or 150 independent climate validations. Calibration subsets rotate across the 36 pairs; the products remain shared. Also, selecting an order retrospectively from the lowest $K_{\mathrm{all}}$ or $N_{\mathrm{mod-cell}}$ requires held-pair stopping depths that are not known ex ante; no prospective cost-minimizing policy is claimed. Full-replay executable verification and its manifest remain in the author-controlled engineering-information archive; the public GitHub companion offers a narrower derived-output verification route.

## 15 Auditable Stage-A stopping decisions on external targets

The original stopping rule is replayed on the unchanged 27-node support. Every source–target run has 36 held comparisons. STC is counted, and a pair is unresolved when the empirical strict-stop condition is never reached. A wrong accepted sign is a *retrospective* finding relative to the full target response; that full response is not an uncertainty-calibrated field measurement. An excess-depth mean is considered comparable only for a fully resolved 36-pair run. Hence a smaller cell count alone cannot certify a valid or efficient laboratory sequence. Table S5 retains the full reporting denominator for wrong accepted signs and unresolved comparisons.

**Table S5:** Independent reporting of the three retrospective decision states, re-aggregated from the archived pair-level replay output. These 38 primary targets are three German system-periods, one Italian full-period sensor and 34 numerically eligible Brazilian plant records; they exclude Safi (separately admitted) and all dependent sensor/year/quarter sensitivities.

| Input family | Runs | Pairs | Wrong | Unresolved |
|---|---|---|---|---|
| Germany (3 periods) | 15 | 540 | 4 | 0 |
| Italy (1 site) | 5 | 180 | 0 | 0 |
| Brazil (34 records) | 170 | 6,120 | 0 | 3 |
| Total | 190 | 6,840 | 4 | 3 |

Of 190 primary historical-source runs, 188 resolve all 36 pairs. The four wrong accepted signs all concern German 2019/System 2B, the same pair (19074-001 versus 19074-008), whose full normalized margin is approximately $-5.3 \times 10^{-5}$; Cocoa, Golden, Switzerland and Albuquerque orders each accept the opposite sign. Eugene leaves two pair comparisons unresolved in Brazil plant PS_009 and one in PS_040. Those three unresolved contrasts have full-response margins of approximately $-5.9 \times 10^{-5}$, $-3.75 \times 10^{-4}$ and $-3.6 \times 10^{-5}$, respectively. They are finite examples of empirical-rule limitations; the fractions 4/6,840 and 3/6,840 are *not* field-instrument error probabilities or population confidence intervals.

The evaluation record sufficient to interpret one Stage-A run contains the target/site/time-window identifier, input sensor/cadence and grid-retention denominators, fixed source order, 36 pair-level stop depths, wrong-accepted and unresolved flags, and the target-local comparison baseline. The independent replay-output checker reconstructs all 900 six-order runs, all pair counts and signs and the 26/26 published numerical assertions. The Safi diagnostic is a separate source/time-synchronization family,

not an added independent plant in this denominator.

## 16 Brazilian plant-level and quarterly limits

Across 51 published plant files, 34 pass the defined simultaneous-source-sample, not-explicit-interpolated, in-grid and 30-date threshold; 17 have no usable joint pair coverage under the registered screen, so the published 51-file corpus is not an analyzed sample of 51 sites. Among those 34, 29 systems have a tracker configuration and five are fixed. Their Brazilian state labels span SP (15), GO (8), RJ (4), MG (3), MS (3), and BA (1). The source release withholds exact station coordinates; plant ID, ownership and meteorological instrument independence cannot be fully verified from those labels. Two plant weight vectors are very similar (total-variation distance 0.0044 for PS_002 and PS_003), so inferential uncertainty across 34 independent climates is inappropriate.

In the 34 primary full-record plant replays all five historical source orders have zero accepted wrong pair signs among *resolved* decisions. The Eugene order does not resolve three held pairs in two plant files; its across-plant depth summaries condition on the remaining 32 fully resolved sites. In 101 eligible within-plant calendar-quarter slices across the corpus, nine wrong accepted signs and fifteen unresolved decisions are observed among the five historical orders. All such subwindows are statistically dependent on their parent plant observations. Among 31 plants with at least two eligible quarter windows, 30 have no source order that is a joint minimum of excess-depth in every available quarter, allowing all exact ties. Figure S2 shows the full qualitative source-minimum record; the author-controlled replay archive retains the exact per-plant/quarter numeric values.

## 17 Trieste single-site annual sensitivity

Nine annual inv1 temperature/POA subsets pass the fixed minimum in-grid count and 30 observed-date gate, with no timezone inference. Their least-burdensome historical order can change by year, and three wrong accepted signs occur across all nine-year × five-order runs. Table S6 reports every year and tied minimum, without suppressing non-optimal or near-tied alternatives.

**Table S6:** Annual subsets of the same Trieste array: in-grid observations, minimum historical-source burden, accepted wrong signs over five orders, and all minimizing sources (A=Albuquerque, G=Golden, S=Switzerland). Not independent sites.

| Year | In grid | Excess (%) | Wrong | Orders |
|---|---|---|---|---|
| 2013 | 9,002 | +1.82 | 0 | A/G/S |
| 2014 | 9,647 | +0.00 | 1 | A |
| 2015 | 10,087 | -6.86 | 0 | A/S |
| 2016 | 9,606 | -5.70 | 1 | A/S |
| 2017 | 9,839 | -5.61 | 0 | S |
| 2018 | 8,953 | -1.44 | 0 | S |
| 2019 | 9,774 | -9.68 | 0 | A/S |
| 2020 | 10,343 | -5.10 | 1 | A |
| 2021 | 9,608 | -2.17 | 0 | S |

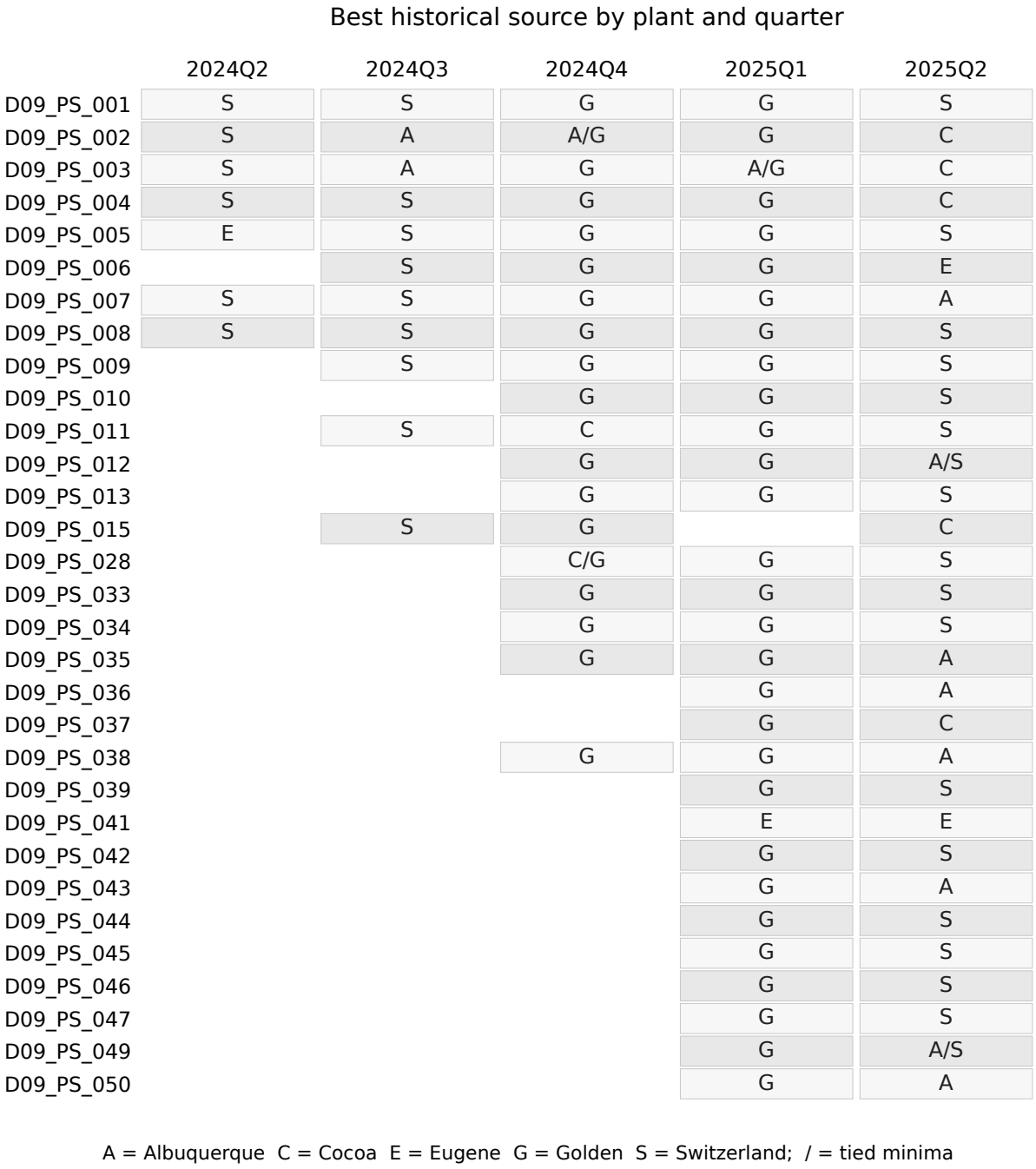



**Figure S2:** Calendar-quarter sensitivity of the frozen five historical source orders within Brazilian plants. Each displayed plant has at least two eligible windows. A, C, E, G and S denote Albuquerque, Cocoa, Eugene, Golden and Switzerland. Slashes denote equal minima; blanks are absent or insufficient quarter support. Rows and quarters are *not* independent-site draws.

## 18 Source-verified out-of-grid geometry diagnostic

The exact frozen 27-node grid spans plane-of-array irradiance 100–1100 $\mathrm{W\,m^{-2}}$ and module temperature 15–75°C, with a sloped lower-temperature edge above 1000 $\mathrm{W\,m^{-2}}$. This descriptive audit reads the original producer files and applies the same positive-irradiance, finite-pair and source-specific screen used for the locked external-target analysis; there is no imputation or extrapolation. Each excluded field record is placed in only one category in this priority order: irradiance below 100, irradiance above 1100, temperature below 15, temperature above 75, and the high-irradiance sloped edge. The categories intentionally do not count overlaps twice. Their sums equal the out-of-grid counts for each source. The source-level breakdown appears in Table S7.

**Table S7:** Source-verified geometric reasons for out-of-grid exclusion. $N$ is valid paired daylight records; $N_{\text{out}}$ is excluded records, and "Low $G$", "Low $T$", "High $G$", "High $T$", and "Edge" are mutually exclusive under the declared priority. Germany is three time-limited system campaigns; Brazil aggregates only the 34 admitted plants. These are not independent climate samples.

| Source | $N$ | $N_{\text{out}}$ | Out (%) | Low $G$ | Low $T$ | High $G$ | High $T$ | Edge |
|---|---|---|---|---|---|---|---|---|
| Switzerland / Task 13 | 34,763 | 6,471 | 18.6 | 5,189 | 1,190 | 92 | 0 | 0 |
| Albuquerque / Sandia | 49,939 | 13,028 | 26.1 | 7,617 | 4,648 | 716 | 0 | 47 |
| Germany (3 campaigns) | 91,285 | 30,479 | 33.4 | 26,344 | 3,916 | 214 | 0 | 5 |
| Italy (1 sensor) | 151,211 | 64,352 | 42.6 | 47,227 | 17,039 | 86 | 0 | 0 |
| Brazil (34 plants) | 398,394 | 128,536 | 32.3 | 108,499 | 4,654 | 11,394 | 3,989 | 0 |

For all 40 inspected target trajectories (one Swiss, one Albuquerque, three German, one Italian and 34 eligible Brazilian plant files), the source-level numbers passing the original support gate match the frozen admission registry. A separately coded analytic convex-polygon membership check agrees with the original Delaunay simplex decision on all finite positive paired samples when a $10^{-8}$ edge tolerance is used for floating-point equality at nominal boundaries. The publisher source-file hashes for Germany, Italy and Brazil are independently matched before processing. The Swiss and Albuquerque files are the originally admitted local source copies.

The diagnostic cannot determine excluded-state outcome performance for the nine response-panel products because no such external-site performance labels were collected. It does not show what the directional burden would have been if the model had been valid outside the original 27-node grid. The original Safi field workbooks are not included in this independent support-subcategory audit; its source-locked inside-grid proportions remain reported separately above, without assigning physical reasons to the excluded Safi records. This audit adds only descriptive admission geometry and does not modify response matrices, weights, test sequences or screening decisions. All supported depth statistics are conditional on the retained and normalized within-grid states; without response profiles for excluded states, neither complete-climate transfer burden nor a bound on its difference from the reported statistics is identified.

**Table S7 reproducibility.** The accompanying reviewer support-audit bundle includes `support_exclusion_audit.py`, the frozen 27-node coordinate file, the frozen admission registry, the derived per-source and family-summary CSVs, a machine-readable validation record and a file-level SHA-256 manifest. The script requires independently obtained original producer files from the cited repositories and reproduces the table without re-running Stage A decisions. The reviewer bundle contains no original third-party raw logger files. The publicly available GitHub Stage-A code/derived-result companion does not include the separately supplied Table S7 reviewer audit bundle; the bundle is provided alongside the manuscript as a supplemental reproducibility file, without producer raw data.

## 19 Retrospective full-campaign workload reinterpretation

For a fixed source order on one target distribution, let $k_{ab}$ be the stopping depth required by the original empirical criterion for distinct products $a, b$ in the nine-product response panel. The main endpoint is the unweighted mean $\overline{k} = 36^{-1} \sum_{a<b} k_{ab}$. For runs in which all 36 pair comparisons resolve, two different planning proxies can be reconstructed without changing any stopping decision:

$$K_{\text{all}} = \max_{a<b} k_{ab}, \tag{S1}$$

$$N_{\text{mod-cell}} = \sum_{a=1}^{9} \max_{b \neq a} k_{ab}. \tag{S2}$$

Equation (S1) defines $K_{\text{all}}$, the longest common prefix when all nine products are exposed to identical conditions in one fixed order. Equation (S2) defines $N_{\text{mod-cell}}$, which assumes each product can stop being measured when all pair comparisons involving it have resolved. It is a hypothetical module–condition observation count, not the measured number of I–V scans or a validated optimum. Neither model prices thermal stabilization, setup changes, preconditioning, repeat sweeps or metrological uncertainty. The 27-node response panel is not a substitute for mandatory IEC characterization.

These two values were reconstructed directly from the 32,400 archived per-pair decision rows across 900 total order–target runs, identifying the 38 primary German/Italian/Brazilian targets and five frozen historical source orders (190 runs, 6,840 pair decisions), and using the target-local order as a reference. Only 188 of 190 historical runs have complete stopping depths. Four accepted-sign discrepancies all relate to one near-tied German comparison; three unresolved pair decisions occur under Eugene in two Brazilian records. These are the previously reported counts, not additional laboratory measurements.

Of the 188 complete historical runs, 79 have a strictly lower mean depth than the local reference. Of these 79, 26 have higher $N_{\mathrm{mod-cell}}$, 40 have higher $K_{\mathrm{all}}$, and 21 increase *both* proxies. The 21 joint-worsening cases cannot improve an additive idealized cost $J_\lambda = K_{\mathrm{all}} + \lambda N_{\mathrm{mod-cell}}$ for any $\lambda \geq 0$; indeed $\Delta J_\lambda = \Delta K_{\mathrm{all}} + \lambda \Delta N_{\mathrm{mod-cell}} > 0$, because both increments are strictly positive. Each $\Delta$ denotes the historical-source value minus its target-local reference value. Here $\lambda$ is an *unspecified ratio* between cost per module–condition and cost per distinct condition; without physical timing logs it is not a measured instrument parameter. Over the full 79 cases the number with $J_\lambda$ worse than local is 40, 42, 41, 35 and 33 at illustrative ratios $\lambda = 0, 0.01, 0.1, 1$, and 10, respectively; these ratios are hypothetical sensitivity points rather than observed lab time ratios. The sets of historical orders tied for minimum mean depth share no member with the minimum $N_{\mathrm{mod-cell}}$ set for 16/38 primary targets, or with the minimum $K_{\mathrm{all}}$ set for 24/38 targets. Exact ties were preserved. The target records are dependent field samples, not independent draws of climates.

For the eligible Brazilian plant record `D09_PS_001`, the local sequence yields $\overline{k} = 285/36 = 7.9167$, $K_{\mathrm{all}} = 14$, and $N_{\mathrm{mod-cell}} = 100$. Replaying the frozen Albuquerque sequence yields $\overline{k} = 258/36 = 7.1667$ ($-9.47\%$), $K_{\mathrm{all}} = 15$ ($+7.14\%$), and $N_{\mathrm{mod-cell}} = 102$ ($+2.00\%$). Both fully resolve the same 36 pairs with no wrong accepted signs. Thus optimizing one per-pair average cannot be represented as an operational guarantee about complete-campaign condition count or measured time. Independent computation is provided in the engineering-application reproducibility package, separate from the locked original replay.

## 20 Restricted matched-physical-unit laboratory/field sensitivity

**Measured grids and the calibration split.** The original NREL characterization workbook (identified by checksum in the reviewer evidence register) contains 22 physical-unit sheets: 20 have an identical grid of 18 positive, source-tabulated corrected $P_{\mathrm{mp}}$ observations, while both `aSiMicro03036` and `aSiMicro03038` lack a complete IEC-related curve. The former is an important Cocoa–Eugene field-yield unit and is *excluded only from this restricted matched laboratory analysis*; no missing value is imputed. The 18 conditions are *not* a complete normative 22-condition IEC 61853-1 matrix: test-equipment limits restricted the temperature to 65°C and left out certain low-temperature cells. This separate experiment does not alter the original nine-product/27-node Stage-A analysis or the full eleven-unit Stage-B field results.

Ten complete sheets represent the same individually identified modules physically redeployed from Cocoa to Eugene; the other ten are distinct Golden modules, not the same serial units. The NREL manual identifies these manufacturer/model correspondences (NREL/TP-5200-61610, 2014, Table 3-1; DOI 10.2172/1130632). The fixed Golden calibration bank consists of `xSi11246`, `mSi460BB`, `mSi0247`, `mSi0251`, `CdTe75669`, `CIGS39013`, `CIGS1-001`, `HIT05662`, `aSiTandem90-31` and `aSiTriple28325`. They are physically disjoint from the ten held Cocoa–Eugene units, although the two groups share manufacturer/model families and the same producer's laboratory archive. Disjoint unit identity does *not* imply independent instrument errors, factories, production batches, or calendar periods.

For each of five source-traced $G_{\mathrm{POA}}$–$T_{\mathrm{mod}}$ trajectories per site, barycentric weights are computed only inside the measured 18-node convex hull, then normalized and averaged equally. The field series are `CIGS39017`, `CdTe75638`, `HIT05667`, `aSiTriple28324` and `xSi12922`; they share their sites and are not independent climate replicates. Valid-daylight support is 134,830/189,706 (71.07%) in Cocoa and 123,104/214,083 (57.50%) in Eugene. The source field yields cover Cocoa in 2011–2012 and Eugene in 2013, while the laboratory characterizations were obtained after their site deployments in April 2014. These data do not isolate a climate-only effect.

**Two explicit calibration information contracts.** For each measured node $j$ define the normalized laboratory response $z_{a,j} = P_{a,j}/P_{a,\mathrm{STC}}$. The source order visits positive-weight, non-STC nodes in descending source-weight order, with STC counted initially. In the *fixed-bank primary sensitivity*, all 45 pairs among the ten Golden calibrators supply full-minus-prefix weighted residual contrasts. These Golden profiles are complete and unchanged for each held target pair; no response from either held

Cocoa–Eugene unit appears inside the calibration residual bank. A pair's acceptance uses the strict inequality between its visited-node partial contrast and the maximum absolute residual from those 45 external calibration pairs. The unvisited full contrast of the held pair is accessed only afterward to label correctness. As a dependent-method sensitivity, the *rotating-bank comparator* instead uses the other eight Cocoa–Eugene modules (28 residual pairs) for each held pair, as previously documented. The two designs use the same 18-node target response data, site weights, two orders and 45 tested pairs, but have materially different calibration prerequisites.

**Table S8:** Ten-unit source-to-target Stage-A replay, 45 held comparisons per direction and calibration design. All four runs per bank resolve all 45 pairs, and no accepted sign disagrees with the retrospective full laboratory contrast. "More/equal/fewer" counts per-pair transfer depth relative to the target-conditioned order. The Golden bank is physically disjoint from test units; neither bank establishes calibrated instrument risk.

| Calibration bank | Transfer | Source mean | Local mean | Excess (%) | More/equal/fewer |
|---|---|---|---|---|---|
| Golden fixed | Cocoa→Eugene | 11.6444 | 9.1333 | +27.4939 | 42/3/0 |
| Golden fixed | Eugene→Cocoa | 12.4222 | 9.6222 | +29.0993 | 45/0/0 |
| Rotating eight | Cocoa→Eugene | 11.0444 | 8.3556 | +32.1809 | 38/7/0 |
| Rotating eight | Eugene→Cocoa | 11.2889 | 9.0889 | +24.2054 | 40/5/0 |

**Workload and conditioning sensitivity.** For the fixed Golden bank, the shared-prefix planning proxy $K_{\text{all}}$ (transfer/local) is 14/13 for Cocoa→Eugene and 17/16 in reverse; the idealized ten-test-module observation proxy $N_{\text{mod}-\text{cell}}$ is 138/122 and 151/132. They *exclude* the 180 complete source-grid calibration observations for the ten Golden modules and physical setup, repeat sweeps and stabilization time. The rotating-bank case likewise requires every source panel profile to be complete across all 45 held-pair comparisons, rather than demonstrating net test-cell savings. The fixed bank does avoid using the tested unit identities in calibration, but no new modules were instrumented prospectively and calibration lifetime or budget has not been evaluated.

Five leave-one-$G$–$T$-trajectory-out replays per direction, removing the same conditioning stream from both sites, keep the Golden bank fixed: transfer excess ranges from +25.42% to +28.88% for Cocoa→Eugene and from +28.18% to +32.08% in reverse. All remain fully resolved with zero laboratory-reference wrong accepted signs. In one Cocoa→Eugene variant, one pair requires fewer cells under the transferred order, so strict per-pair dominance is *not* a general claim. Pooling the observed records rather than equally averaging the five source streams reproduces the primary percentages at the displayed precision. These source-composition checks are dependent deterministic perturbations, not independent climate observations or coverage probabilities.

**Separate Stage B and endpoint mismatch.** The same ten *test-unit* identifiers are screened using the previously source-traced published annual final yields, with site-best denominators recalculated over exactly ten and $q = 5$ percentage points as a nominal decision resolution. Within this subcohort `CdTe75638` is the field-yield point leader at both sites, as well as the leader in the separately weighted normalized laboratory scores; these quantities have different physical definitions. The full original eleven-unit field screening result (three of five excluded Cocoa→Eugene) is unchanged.

**Table S9:** Ten-unit Stage-B field-yield shortlist, distinct from the laboratory Stage-A score. At $q = 5\%$, the Cocoa→Eugene exclusion of `HIT05667` occurs over the nominal resolution interval $[3.239, 6.596)$ percentage points; no uncertainty-certified exclusion is claimed.

| Source → target | Target set | Missing | Excluded unit | Coverage |
|---|---|---|---|---|
| Cocoa → Eugene | 4 | 1 | `HIT05667` | 3/4 (75%) |
| Eugene → Cocoa | 3 | 0 | None | 3/3 (100%) |

**Calibration-bank subset sensitivity.** We exhaustively evaluated the $\binom{10}{8} + \binom{10}{9} + \binom{10}{10} = 56$ fixed Golden-bank subsets of size 8–10. For each subset, the 45 held test pairs and two target climates were unchanged. A separate vectorized evaluator first reproduced the 180 full-bank reference decisions; it then reproduced 10,080 subset-based decisions (56 banks × 4 source/target runs × 45 pairs), all resolved without wrong accepted laboratory-reference signs. The mean transferred-depth penalty remained positive in both directions, but fell to +5.51% for Cocoa→Eugene after removing `xSi11246`

and CIGS39013. These heavily overlapping replays measure sensitivity to library composition, not independent calibration experiments or a risk probability.

**Table S10:** Calibration-bank sensitivity on the unchanged ten test units, 18-node grid and site weights. "Wrong" refers to accepted signs versus the complete laboratory reference. All bank subsets of size 8–10 overlap and include the 10 leave-one-out cases; the model-exclusion design selects calibrators per held pair.

| Calibration restriction | Banks | Cocoa→Eugene | Eugene→Cocoa | Wrong / all pairs |
|---|---|---|---|---|
| Full Golden bank (ten profiles) | 1 | +27.49% | +29.10% | 0/180 |
| Exclude one Golden profile at a time | 10 | +13.87 to +29.70% | +28.87 to +31.54% | 0/1,800 |
| Exclude same model as each held pair | 1 design | +25.63% | +30.36% | 0/180 |
| All fixed Golden subsets of size 8–10 | 56 | +5.51 to +30.35% | +26.55 to +31.54% | 0/10,080 |

**Envelope-containment limit.** With the full Golden bank and Cocoa→Eugene transfer, two held-pair residuals exceed the bank's maximum residual at stopping: CIGS39017/mSi460A8 (0.00370685 versus 0.00358310) and aSiTandem72-46/xSi12922 ($8.63379 \times 10^{-5}$ versus $8.01512 \times 10^{-5}$). Both accepted signs nevertheless match their full laboratory references. The pair-specific same-model-exclusion variant also exhibits containment breaches. The empirical Golden envelope is therefore not a deterministic bound for held units and supplies no calibrated decision guarantee.

Four of the 45 laboratory full-score pair signs differ from the field annual-yield pair signs in Cocoa and four in Eugene. These are differences between physical endpoints and measurement dates, *not* predictions that the laboratory accepted a wrong field module. Neither the illustrative $q = 5$ cutoff nor the source-reported $Y_f$ uncertainty establishes metrologically certain shortlist membership. In particular, the target-best normalizing denominator and covariance remain uncharacterized.

**Reproducibility and scope.** Separate direct and vectorized implementations reproduce all 180 fixed-bank pair decisions; the earlier rotating-bank replay also matches 180/180. Recomputing weights from the frozen reduced source trajectories matches 36/36 aggregated weight entries, but the publicly shareable aggregate-weight route alone does not recreate raw field telemetry. Original producer sheets and logger files are not redistributed. Unit-identity separation strengthens the within-source holdout; measurement uncertainty, noncontemporaneous source history, incomplete grid support and endpoint mismatch still preclude a prospective two-stage certificate.

**Three-state sign logic.** A full-reference contrast can in principle be positive, negative, or exactly zero. The historical Golden auditor used a Boolean comparison of "greater than zero" and would therefore treat a negative accepted sign against an exact zero full contrast as if both were nonpositive. The audit logic is corrected conceptually to

$$\operatorname{sgn}_3(x) = \begin{cases} -1, & x < 0, \\ 0, & x = 0, \\ +1, & x > 0, \end{cases}$$

with an accepted nonzero sign against a full-reference tie reported as a separate *accepted-against-reference-tie* state rather than a correct sign. The archived external 32,400 pair records contain no exact full-reference ties, and the 180 fixed-Golden pair decisions contain none; the minimum absolute Golden full contrast is approximately $9.45 \times 10^{-5}$. A tie-aware replay changes no historical decision or reported result. This correction is therefore a code-logic boundary for future reuse, not an empirical revision of the present study.

## 21 Fixed-bank order baselines and local-heuristic stress

With the same ten tested units, ten fixed Golden calibrators, 18 measured nodes and two target weight vectors, we evaluate 5,000 seeded random permutations of 17 non-STC nodes per target (NumPy PCG64 default_rng, seeds 20261106 and 20261107). We also enumerate the $\binom{17}{2} = 136$ single transpositions of each destination-weight-descending order. Each order is scored on all 45 held pairs against the same 45 Golden calibration-pair residuals. No products, climate windows or source records are resampled.

**Table S11:** Fixed-bank order diagnostics; means count STC. All random and single-swap runs resolve 45/45 pairs with no wrong accepted laboratory-reference signs. "Random ≤ transfer" counts random orders at least as economical as transferred; none is at least as economical as local. Best swaps are retrospective oracles, not prospective policies.

| Target | Local | Transferred | Random min/median | Random ≤ transfer | Best one-swap | Improving swaps |
|---|---|---|---|---|---|---|
| Cocoa | 9.6222 | 12.4222 | 10.2444/15.1333 | 195/5,000 | 8.9556 | 15/136 |
| Eugene | 9.1333 | 11.6444 | 10.0889/15.2222 | 40/5,000 | 8.6889 | 11/136 |

The transferred orders outperform most sampled random permutations, but the local heuristic outperforms all sampled orders. Nevertheless, 15/136 Cocoa swaps and 11/136 Eugene swaps are better than local; another 18 swaps tie at each site. This demonstrates nonoptimality in the one-swap neighborhood, not a new deployable ordering algorithm: scoring any improved swap uses the held products' completed laboratory responses. These order perturbations are dependent diagnostics, not independent climates or evidence of metrological decision-risk coverage.

The accompanying reviewer code bundle stores all order scores and seeds. A separate scalar implementation verifies 1,260 selected pair decisions exactly. Original producer power sheets are not redistributed.

## 22 Decision-task decomposition and three-state sign audit

The task-specific analysis is reconstructed from the same pair-level stopping records; it does not alter the stopping rule or the response data. For a fully resolved run, the pairwise mean, shared-prefix depth $K_{\mathrm{all}}$, and module–condition proxy $N_{\mathrm{mod-cell}}$ are defined in the Main. Winner identification is treated separately: accepted pairwise signs induce directed wins, and a unique product with wins over all $m-1$ alternatives is the accepted winner. Its accepted-winner depth $K_{\mathrm{win}}$ is the latest stopping depth among those incident wins. The full weighted-score winner is used only after stopping to label the certificate correct or wrong.

Table S12 shows that the portability conclusion can reverse even on the original nine-product panel. Cocoa→Eugene is worse than Eugene's target-conditioned heuristic for the all-pairs mean and both campaign proxies, but better for accepted-winner depth. Golden→Eugene has the same direction of mean/winner disagreement. Eugene→Cocoa is worse under every displayed metric. These are finite deterministic contrasts on the same response panel, not independent experiments.

**Table S12:** Decision-task metrics on the original three U.S. target environments. Depths count STC. $K_{\mathrm{win}}$ is reported only for the unique accepted winner; every row shown has the correct full-reference winner.

| Target | Source order | $\bar{k}$ | $K_{\mathrm{all}}$ | $N_{\mathrm{mod-cell}}$ | $K_{\mathrm{win}}$ |
|---|---|---|---|---|---|
| Cocoa | Cocoa | 8.222 | 15 | 110 | 5 |
| Cocoa | Eugene | 12.000 | 20 | 163 | 9 |
| Cocoa | Golden | 8.083 | 17 | 115 | 6 |
| Eugene | Cocoa | 8.750 | 17 | 133 | 8 |
| Eugene | Eugene | 8.083 | 14 | 118 | 14 |
| Eugene | Golden | 8.250 | 14 | 116 | 8 |
| Golden | Cocoa | 10.056 | 18 | 130 | 5 |
| Golden | Eugene | 14.417 | 23 | 179 | 13 |
| Golden | Golden | 9.139 | 20 | 127 | 6 |

Across the 38 primary external target distributions, all 190 historical source-order runs produce a unique accepted winner and 186 of those winners match the full weighted-score winner. The four wrong winners are the already reported 2019/System 2B near-tie under Cocoa, Golden, Switzerland and Albuquerque orders. Two of the 186 correct-winner runs (Eugene on Brazil PS_009 and PS_040) still contain unresolved non-winner pairs, so an all-pairs mean is unavailable. Among the remaining 184 correct and all-pairs-complete runs, the pair-mean/winner-depth direction counts are: 59 better/better, 17 better/same, 3 better/worse; 2 same/better; and 9 worse/better, 25 worse/same, 69 worse/worse. Hence 12 runs have a strict sign reversal between pair-mean change and winner-depth change. This denominator distinction is necessary: winner identification can finish even when unrelated non-winner pairs remain unresolved. As a check against hindsight-conditioned reporting, all 188 all-pairs-complete

historical runs yield 16 raw strict direction reversals, but four of these are incorrect accepted winners, all from the same 2019/System 2B target (Cocoa, Golden, Switzerland and Albuquerque orders). These four appear to improve winner depth while worsening pairwise mean, but are *invalid* winner-identification successes. The 12 correctly identified cases therefore remain the appropriate validity-gated retrospective comparison. This restriction does not turn the empirical stopping envelope into a prospective error bound. For only two products, all relevant depths equal the single pair depth, so such strict depth reversals require at least three products.

Among the 79 historical orders whose complete all-pairs mean is lower than the corresponding target-local heuristic, 40 have a larger $K_{\text{all}}$, 26 have a larger $N_{\text{mod-cell}}$, and 21 have both larger. Winner depth is better in 59 of these 79, unchanged in 17, and worse in 3. These categories reuse the same nine product matrices and correlated field targets; they are a finite task-discordance ledger, not binomial observations.

The fixed-Golden ten-test-unit sensitivity provides a counterexample to any claim that winner identification is generally easier to transfer. Table S13 shows that both transfer directions worsen pair-average, campaign, and winner-certificate depth. The appropriate conclusion is therefore *task-indexed portability*, not a universal ordering between tasks.

**Table S13:** Task metrics for the restricted ten-test-unit, fixed-Golden-bank sensitivity. Every row resolves all 45 held pairs and identifies the correct full-laboratory winner.

| Target | Source order | $\bar{k}$ | $K_{\text{all}}$ | $N_{\text{mod-cell}}$ | $K_{\text{win}}$ |
|---|---|---|---|---|---|
| Cocoa | Cocoa | 9.622 | 16 | 132 | 10 |
| Cocoa | Eugene | 12.422 | 17 | 151 | 12 |
| Eugene | Eugene | 9.133 | 13 | 122 | 9 |
| Eugene | Cocoa | 11.644 | 14 | 138 | 13 |

## 23 Magnitude and source-family concentration of strict task reversals

The twelve retrospectively valid strict reversals (Table S14) are the same cases counted in the Main, not additional trials or independent replications. Nine rows belong to the Brazilian plant-record family; the 12 rows represent 11 distinct target records because one Brazilian target occurs under two transferred orders. In nine rows, the accepted-winner depth changes by only one visited test cell; every absolute percentage change in mean pairwise depth is below 5%. The full-reference correctness condition is applied after the replay, not at stopping time. The signs and magnitudes are descriptive of this source-locked panel and do not identify a population distribution of effects, laboratory-time savings or a climate-only causal mechanism.

**Table S14:** All 12 retrospectively correct, all-pairs-complete strict task reversals from the frozen R16 decision ledger. Deltas are transferred source order minus target-local heuristic; a positive mean delta denotes extra pairwise burden and a negative winner-depth delta denotes earlier accepted-winner identification. All 12 accepted winners match the complete reference. Repeated runs share the same nine response profiles and field-source families.

| Target | Source order | $\Delta\bar{k}$ (%) | $\Delta K_{\text{win}}$ | $\Delta K_{\text{all}}$ | $\Delta N_{\text{mod-cell}}$ |
|---|---|---|---|---|---|
| DE:2018_System_1 | Golden | +1.70 | -1 | -1 | +2 |
| DE:2019_System_2B | Eugene | +3.76 | -1 | 0 | +8 |
| IT:Trieste_inv1_tpan | Cocoa | +4.86 | -6 | +3 | +7 |
| BR:PS_006 | Albuquerque | -1.50 | +2 | +1 | +9 |
| BR:PS_008 | Switzerland | +2.61 | -1 | 0 | +10 |
| BR:PS_008 | Albuquerque | +0.98 | -2 | +2 | +18 |
| BR:PS_010 | Cocoa | +3.52 | -1 | +3 | +20 |
| BR:PS_034 | Albuquerque | +4.83 | -1 | +6 | +22 |
| BR:PS_037 | Cocoa | +1.23 | -1 | +2 | +8 |
| BR:PS_043 | Golden | -3.80 | +1 | -1 | -8 |
| BR:PS_049 | Cocoa | +1.93 | -1 | -1 | +10 |
| BR:PS_050 | Cocoa | -2.34 | +1 | -3 | -12 |

*BR/IT/DE denote Brazil, Italy and Germany; full source IDs and precision are retained in the companion CSV.*

## 24 Preference-space robustness and illustrative fuzzy $\alpha$-cuts

**Scope and inputs.** This section adds an *interpretive* analysis of the frozen 190-run R16 comparison ledger and the same eleven published Cocoa–Eugene yields in Table S2. It does not alter any Stage-A stopping depth, response matrix, winner,

stage assignment, measured power, or Stage-B source/target yield. The analysis distinguishes uncertainty over a *user's decision priorities* from unmeasured physical uncertainties, which cannot be supplied by fuzzy membership alone. The source SHA-256 checks and two independently executed reproduction paths are available both in the separate R20 reviewer reproducibility package and at the https://github.com/shaikhamalkawi-ux/pv-evidence-portability-across-climates/tree/c7455d83f0d84738c55c265ab53feaf526ccb8de/reproducibility/R20 archive.

## Exact campaign-weight dominance before choosing a fuzzy membership

For every correct-winner, all-pairs-complete historical replay whose pairwise mean depth improves on the target-local order, define

$$\Delta J_\lambda = \Delta K_{\text{all}} + \lambda\, \Delta N_{\text{mod}-\text{cell}}, \qquad \lambda > 0.$$

These are *source-minus-local* differences in the two previously specified, hypothetical campaign proxies. Negative $\Delta J_\lambda$ means the transferred sequence has a smaller combined burden under that weight. Because this function is affine, strict improvement for every $\lambda > 0$ occurs precisely when both component differences are nonpositive and at least one is negative; strict worsening for every $\lambda > 0$ has the reverse condition. Opposite-signed differences cause one positive transition at $\lambda^* = -\Delta K_{\text{all}}/\Delta N_{\text{mod}-\text{cell}}$. Among 79 eligible mean-improving replays there are **34 always-better**, **30 always-worse** and **15 weight-sensitive** cases. The 30 always-worse cases comprise 21 strictly worse on both proxies and nine worse on one proxy with the other unchanged. These are 79 dependent computational replays, not independent target climates. The 15 mixed-sign runs have eight distinct breakpoints: 1/14, 1/4, 2/5, 1/2, 2/3, 1, 3 and 4. The complete target identifiers, exact deltas and critical ratios are supplied in the derived CSV.

A concrete independent-of-fuzziness example is Brazil `D09_PS_001` under Albuquerque: $\Delta\bar{k} < 0$, but $\Delta K_{\text{all}} = +1$ and $\Delta N_{\text{mod}-\text{cell}} = +2$, so $\Delta J_\lambda = 1 + 2\lambda > 0$ for any $\lambda > 0$. This strictly worsened illustrative campaign burden is not rescued by any positive fuzzy weighting of the two proxies.

## Illustrative triangular possibility profile for a relative campaign weight

An analyst-declared—not measured or elicited—triangular profile $\widetilde{\lambda} = (0.05, 1, 10)$ spans all eight observed breakpoints. Its $\alpha$-cut is

$$[\lambda_L(\alpha), \lambda_U(\alpha)] = [0.05 + 0.95\alpha,\ 10 - 9\alpha], \quad 0 \le \alpha \le 1.$$

The minimum and maximum $\Delta J$ over a cut occur at the interval endpoints, so no fuzzy optimization heuristic is needed. Table S15 reports exact signs: "both" means that positive and negative values are attainable in the declared cut; an equality at the point core is reported separately and *not* called a robust win. In particular, the full-support 34/30/15 split is independent of the choice of the triangle's modal point, while the higher-$\alpha$ splits are not.

**Table S15:** Illustrative campaign-weight $\alpha$-cut outcome for the 79 dependent mean-improving historical replays, with $\widetilde{\lambda} = (0.05, 1, 10)$. Counts are possibilities under a stipulated preference membership, not frequencies or error probabilities.

| $\alpha$ | Relative-weight interval | Robustly better | Robustly worse | Both signs | Exact tie |
|---|---|---|---|---|---|
| 0 | $[0.05, 10]$ | 34 | 30 | 15 | 0 |
| 0.5 | $[0.525, 5.5]$ | 41 | 32 | 6 | 0 |
| 0.75 | $[0.7625, 3.25]$ | 42 | 34 | 3 | 0 |
| 1 | $\{1\}$ | 42 | 35 | 0 | 2 |

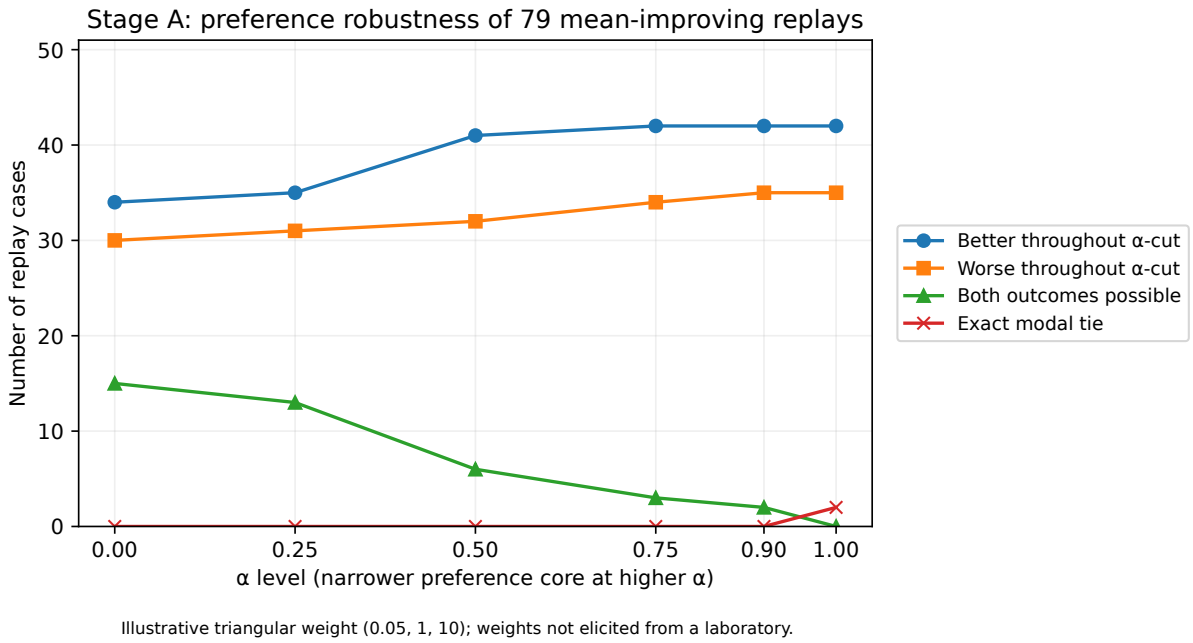



**Figure S3:** Illustrative campaign preference ambiguity with the modal weight fixed at 1. The source profiles, stopping decisions and response values are untouched. These $\alpha$-cut counts report sensitivity to a stipulated preference shape, not a distribution of laboratory costs.

**Membership falsification.** Holding the support $[0.05, 10]$ fixed but moving the modal weight from 0.25 to 1 to 4 changes the number of both-sign cases at $\alpha = 0.5$ from 14 to 6 to 3. Hence the apparently stronger conclusions within narrow fuzzy cuts depend on subjective membership choices. The correct contribution is the exact breakpoint analysis, not an estimated physical possibility that a particular order saves time. This approach is consistent with the possibility-versus-probability distinction and exact $\alpha$-cut representation cited in the Main (References 40–41), without claiming novelty for fuzzy arithmetic.

## Illustrative tolerance-priority profile for shortlist exclusion

We separately retain the exact eleven physical-module annual yields in Table S2 and recompute both source and target candidate sets for each cutoff $q$. No yield is fuzzified. For the analyst-declared $\widetilde{q} = (4.5\%, 5\%, 5.5\%)$ the nominal Cocoa-to-Eugene false-exclusion count equals three at its modal $q = 5\%$, but the whole support allows two to four. Table S16 and Figure S4 show the exact interval boundaries: the original `aSiTandem72-46` omission ceases as its Cocoa deficit is reached (5.0520059%), while `xSi12922` and `mSi460A8` become newly target-relevant at 5.1282051% and 5.3306343%, respectively. Two nominal omissions, `HIT05667` and `aSiTriple28324`, persist across the chosen preference interval. Set $b_1$, $b_2$ and $b_3$ to the *exact* respective deficit thresholds for these three changes (approximately 5.05200594, 5.12820513 and 5.33063428 percentage points); the rounded displays below do not define the interval endpoints. This does not establish metrological robustness of their observed yields.

**Table S16:** Exact nominal Cocoa-to-Eugene false-exclusion count over a declared preference interval in $q$. Boundaries are from the original eleven annual yields; intervals are left-closed and right-open except the final shown endpoint.

| Tolerance $q$ (percentage points) | Excluded target-relevant units |
|---|---|
| $[4.5, b_1)$ | 3 |
| $[b_1, b_2)$ | 2 |
| $[b_2, b_3)$ | 3 |
| $[b_3, 5.5]$ | 4 |

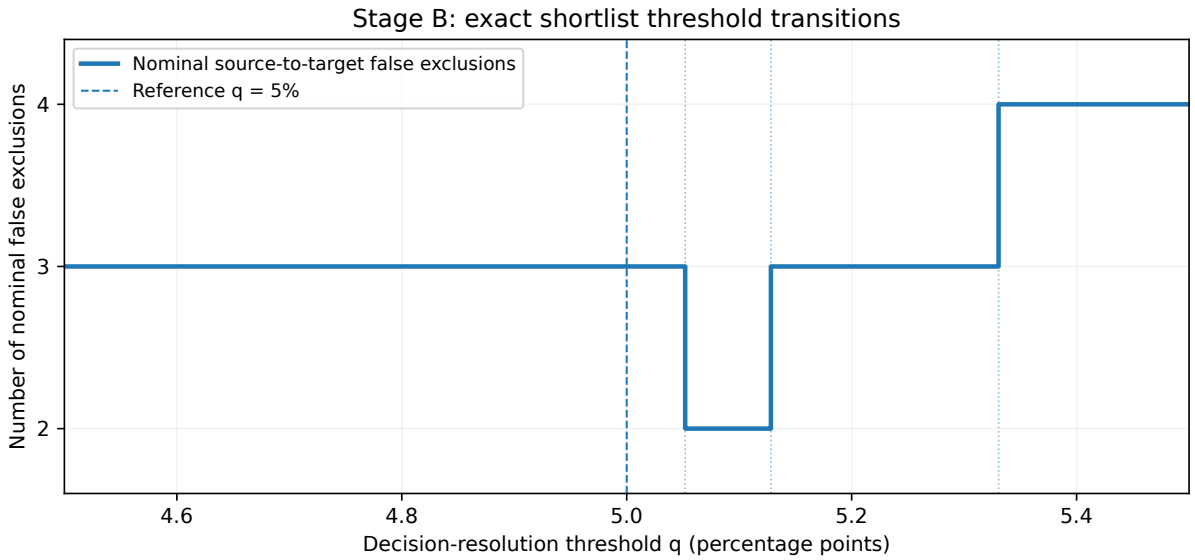



**Figure S4:** Stage-B shortlist sensitivity around $q = 5\%$, recomputed from the same eleven original yields at every threshold. The triangular profile represents a hypothetical tolerance preference, not uncertainty in measured final yield or a calibrated false-exclusion probability.

**Interpretation boundary.** Under the stipulated $q$ membership, two units have possibility and necessity grades equal to one for nominal exclusion; the close-to-boundary `aSiTandem72-46` has necessity about 0.104. Newly possible exclusions `xSi12922` and `mSi460A8` have possibility grades about 0.744 and 0.339. Such grades quantify the chosen *preference profile*, not the occurrence probability or physical uncertainty of any unit. No instrument measurement, calibration-uncertainty propagation, time savings, independent site validation or prospective screening claim follows from these illustrative calculations.

# 25 Graph-bottleneck attribution of task reversals

**Purpose and data restriction.** This section explains the 12 already reported, retrospectively valid task reversals (Table S14) from the original 32,400 derived pair-decision records. It does not change response matrices, accepted signs, visited test cells, target weights, or the original stopping rule. Results here are deterministic explanations of the *same* nine products and correlated target histories, not a new physical or independent-site experiment.

**Comparison graph and completion bottlenecks.** Represent the nine products by vertices $V$ of a complete undirected graph. Each edge $e = \{a, b\}$ carries the archived pair stopping depth $k_e(\pi)$ under fixed measurement order $\pi$. Condition on every pair resolving and on a unique accepted winner $w$ that matches the full-reference winner *after* the replay. The eight incident edges $S_w = \{\{w, b\} : b \neq w\}$ constitute the winner star; the remaining 28 edges $L_w$ join non-winners. At depth $k$, the graph of resolved pairs contains exactly those edges with $k_e(\pi) \leq k$. Consequently the earliest depth at which all eight correctly oriented winner edges are resolved and the depth at which every edge is resolved are

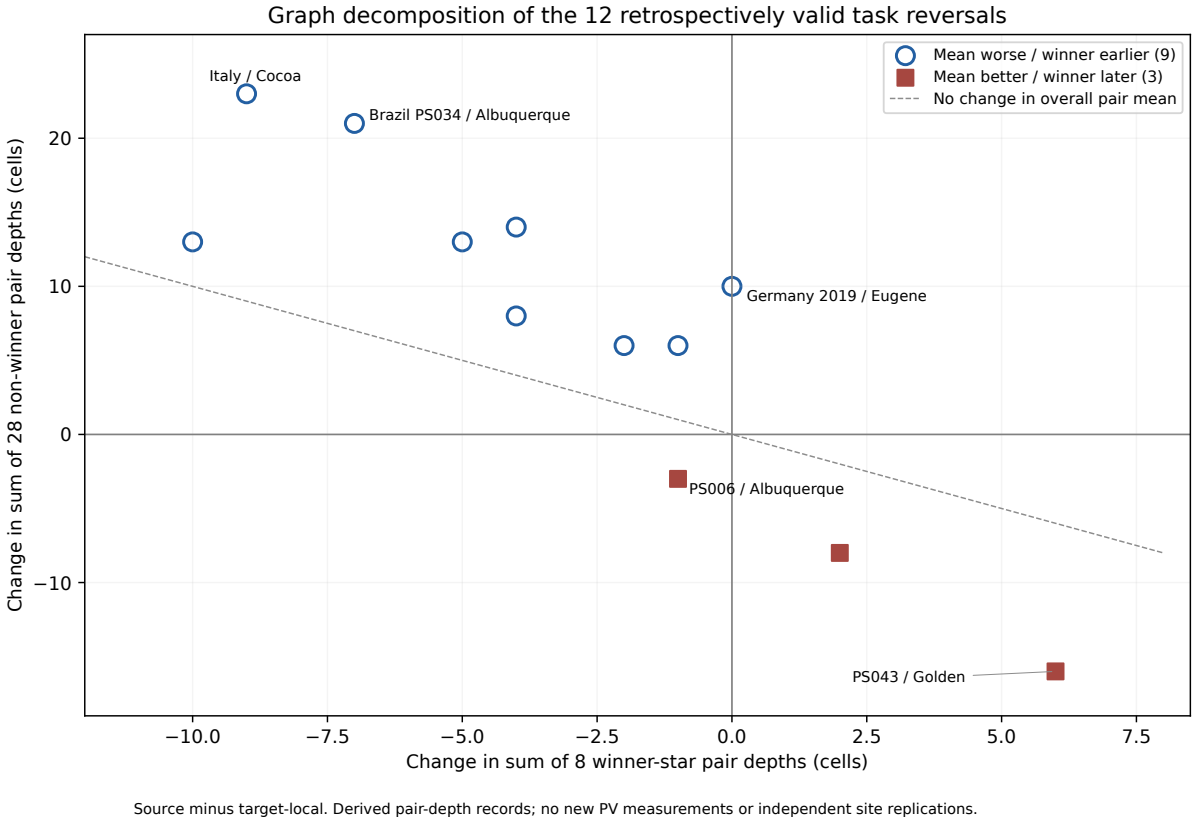
$$K_{\mathrm{win}}(\pi) = \max_{e \in S_w} k_e(\pi), \qquad K_{\mathrm{all}}(\pi) = \max_{e \in S_w \cup L_w} k_e(\pi).$$

The mean over all 36 pairs decomposes exactly as

$$\bar{k}(\pi) = \frac{8\,\bar{k}_{S_w}(\pi) + 28\,\bar{k}_{L_w}(\pi)}{36}, \qquad 36\,\Delta\bar{k} = \sum_{e \in S_w} \Delta k_e + \sum_{e \in L_w} \Delta k_e,$$

where $\Delta$ means the historical source order minus the target-local heuristic on the *same* target and with the same retrospectively correct winner. The mean uses every edge; the winner task uses a maximum over just the winner-star edges. These distinct operations explain why the signs of $\Delta\bar{k}$ and $\Delta K_{\mathrm{win}}$ need not agree. If all 36 edge depths of one order are componentwise no larger than those of another, all four previously defined depth/campaign proxies are non-increasing, *provided* the accepted-winner identity is correct and shared. These graph identities are elementary mathematical interpretations, not novel PV-specific theorems.

**What the frozen decision record shows.** Reconstruction from 900 target/order groups and 32,400 archived pair rows reproduces 188 complete historical-order runs, of which 184 also identify the correct winner; 12 are strict pair-mean/winner-depth reversals. In all nine cases where the transferred order worsens the mean yet identifies the correct winner sooner, the summed depths of the 28 non-winner edges increase while the summed eight winner-star depths do not increase. In the other three cases, where the mean improves but winner identification is later, the non-winner-edge sum decreases. These sums explain which edges contribute to the mean but do *not* replace the bottleneck maximum: one winner-star edge can become slower even when the winner-star sum decreases. The table behind this figure records all 12 target/order identifiers and the individual aggregate changes; it does not treat them as 12 independent climates.




Source minus target-local. Derived pair-depth records; no new PV measurements or independent site replications.

**Figure S5:** Exact graph attribution of the 12 correct-winner, all-pairs-complete strict task reversals using frozen pairwise stopping depths. The winner-star comprises eight edges; all other comparisons comprise 28 edges. Their depth-sum changes explain the change in pairwise mean, whereas winner identification is governed by the *maximum* winner-star depth. These are repeated retrospective replays, not new PV measurements or independent-site replicates.

**Two source-locked examples.** For the Trieste primary record under the transferred Cocoa order, winner-star edge depths decrease in sum by 9 cells but the remaining edge depths increase in sum by 23. Thus $\Delta\bar{k} = (-9+23)/36 = +0.3889$ cells, even though the correct winner is identified at depth 8 rather than 14; full-campaign completion rises from 14 to 17. For Brazilian target `D09_PS_006` under Albuquerque, the winner-star sum changes by $-1$ and the other-edge sum by $-3$, so the mean improves by $4/36 = 0.1111$ cell; nonetheless the slowest winner-star comparison rises from 9 to 11 and delays identification. In both examples, the compared sequence and local reference share the retrospectively verified full-response winner.

**Secondary Pareto interpretation and limits.** Comparing the six available orders on the four-dimensional vector $(\bar{k}, K_{\mathrm{win}}, K_{\mathrm{all}}, N_{\mathrm{mod-cell}})$, after excluding wrong-winner or incomplete outcomes, gives 222 eligible target/order records. Twenty-three of 38 targets have at least two nondominated tested orders; the local weight-descending heuristic is dominated by a historical order in 16 targets. This is only a *within-tested-orders, post-hoc* comparison: choosing a sequence on these outcomes presupposes full retrospective depths and correctness, so it neither identifies a global optimum nor supplies a deployable future rule. Independent reconstruction verifies the underlying pair-decision and four-metric totals; the reproducibility bundle contains no additional original third-party raw data. The paired-depth ledger, graph attribution and independent integer checks are available at the fixed public GitHub snapshot https://github.com/shaikhamalkawi-ux/pv-evidence-portability-across-climates/tree/14c8508adb9dfc62cb6bc51975832754fa5973c1/reproducibility/R21.